K-MEANS CLUSTERING SOLUTION (k = 2)

1. Clustering vs Classification

- Classification is supervised learning using labeled data.

- Clustering is unsupervised learning that groups points by similarity without labels.

2. Data Points

a (4,2), b (3,3), c (4,3), d (5,3), e (1,2), f (2,1), g (1,1), h (3,2)

3. Initialization

Initial centroids chosen:

C0 = (4,2)

C1 = (1,1)

4. Pass 1 – Assign Points to Nearest Centroid

Cluster 0: a, b, c, d, h

Cluster 1: e, f, g

5. Recompute Centroids

C0 = (3.8, 2.6)

C1 = (1.3333, 1.3333)

6. Compute WCV (Within-Cluster Variation)

Cluster 0 sum of squared distances = 4.0

Cluster 1 sum of squared distances = 1.3333

WCV1 = 5.3333

7. Compute BCV (Between-Cluster Variation)

Global mean = (2.875, 2.125)

BCV1 = 14.4174

8. Ratio = BCV / WCV

Ratio1 = 2.7031

9. Pass 2 – Reassign Points

All assignments unchanged. Algorithm converged.

10. Final Clusters

Cluster 0: a, b, c, d, h

Cluster 1: e, f, g

Final WCV = 5.3333

Final BCV = 14.4174

Final Ratio = 2.7031

The ratio increased from the initial pass, showing convergence and improvement of the k-means objective.